

Fischer–Tropsch synthesis: activation of low-alpha potassium promoted iron catalysts

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Abstract

The purpose of this study is to investigate the influence of activation gas type (CO, syngas and H₂), activation temperature and pressure on FTS activity and selectivity. A series of Fischer–Tropsch synthesis (FTS) reactions in a continuously stirred tank reactor (CSTR) for 400–1000 h were carried using a Fe/K/Si = 100/1.25/5.1 (atomic ratio) catalyst. The highest CO conversion was obtained for a CO activated catalyst while hydrogen activation yielded the lowest, regardless of the activation temperature. A higher activation temperature yielded a higher peak CO conversion during the induction period. FTS activity for syngas pretreatment showed a dependency on the activation pressure. In addition to the FTS activity, CO activation showed the best overall hydrocarbon productivity. The lowest hydrocarbon production rate was obtained with the H₂ activated catalyst. Carbon monoxide activation also yielded the lowest methane selectivity and activation temperature did not affect the CH₄ selectivity in the range of 230–270 °C. Activation temperature had little effect on the formation of olefin (C₂–C₂₀) and the olefin fraction decreased in the order of H₂>syngas>CO activation. The induction period was present following each of the three pretreatments: CO, H₂, or syngas with the same H₂/CO ratio as used during the synthesis period.

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1. Introduction

In the Fischer–Tropsch Synthesis (FTS) process, the hydrocarbon is produced from CO and H₂ by a reaction that can be expressed as:



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When an iron catalyst is used for FTS, the water gas shift (WGS) reaction can also occur. This reaction consumes CO and water formed by the FTS reaction and produces additional hydrogen as well as carbon dioxide.



Potassium has long been used as a promoter for iron catalysts. It provides an increase in the alkene yield and a decrease in the CH_4 selectivity [1–3]. Potassium can also increase the catalytic activity for FTS and water gas shift (WGS) reactions [3–9].

Pretreatment of iron catalysts can have important influences on the FTS activity and selectivity. A cobalt catalyst is usually activated with H_2 and metallic cobalt is believed to be the active phase for FTS, but the pretreatment of iron catalyst is not as simple as that of a cobalt catalyst. Carbides are probably the more active phase for FTS. Table 1 shows the phase changes in an iron pretreatment process [1]. Anderson [10] pointed out that without pretreatment the iron catalyst is inactive, and H_2 , CO, or $\text{H}_2 + \text{CO}$ can reduce the catalyst at 200–300 °C. He found that at 0.1 MPa and 325 °C, pure CO was a better reduction gas than syngas (1 H_2 + 1.5 CO). Based on a 50% gas contraction ratio (without CO_2), a catalyst following CO pretreatment can maintain service life for 350 days while a syngas pretreated material had a life of only 200 days. Baltrus et al. [11] applied XPS, Auger electron spectroscopy (AES) and ion spectrometry to study the state of surface iron and carbon and the distribution of K and Cu over a Fe/K/Cu catalyst. They found that the surface K/Fe ratio from $\text{CO} + \text{H}_2$ reduction was greater than that from CO reduction; therefore, $\text{CO} + \text{H}_2$ reduction resulted in a more inactive carbon deposit which covered part of the active carbide surface.

Sault [12] studied the activation effect of CO or H_2 on Fe/K/Cu = 100/0.2/3 (wt.%) and Fe/K/Cu/Si = 100/4.2/5/25 (wt.%) catalysts using AES. He found that CO conversion over a Fe/K/Cu = 100/0.2/3 (wt.%) catalyst can vary by a factor of nearly 3, depending on the activation conditions. H_2 activation leads to the rapid transformation of iron from Fe_2O_3 to metallic iron and segregation of sulfur to the surface. Sulfur arises from bulk sulfate impurities present in the nitrate salt in catalyst preparation. Sulfur coverage increased with activation time and temperature since the sulfur diffusion rate increases as the temperature increases. CO conversion over a Fe/K/Cu/Si = 100/4.2/5/25 catalyst changed to a much lesser extent with activation conditions. No significant surface composition change was observed over the iron catalyst with a higher K content. He believed that this result was due to the difference in surface area and the reducibility of the two catalysts. He also found that at a high temperature, sulfur can be segregated to the surface during activation. Silica is a structural promoter added to maintain the surface area [13–15] and it can also have a chemical effect on catalyst properties [14,16–18]. K facilitates CO dissociation and carbon

Table 1
Degree of reduction at 673 K and 8 h at fixed space velocity [1]

Gas	Reduction (%) [*]	Phase present
100% H_2	80	$\alpha\text{-Fe}$, Fe_3O_4
98% H_2 + 2% CO	20	Fe_5C_2 , Fe_3O_4
100% CO	4	Fe_5C_2 , Fe_3O_4

^{*} Amount of iron oxide converted to carbide or metallic forms.

chain growth [1,10,15,19–22]. It has been reported that both K [16,17] and Si [14,16–18] can inhibit the reduction of iron.

Reduction of some or all Fe in Fe_3O_4 with H_2 to zero valence was claimed by some researchers [23–30]. It is found that 20% metallic iron was obtained by treating an iron catalyst with H_2 at 220 °C. Pretreatment of an iron catalyst with CO or syngas can result in the formation of χ -carbide [26–28]. Bukur studied the activation effect on a K/Fe = 0.8/100 (mass) catalyst in a fixed bed reactor and concluded that CO activation at 280 °C led to higher initial FTS activity and a higher fraction of long chain hydrocarbons than an H_2 activated catalyst.

2. Experimental

2.1. Preparation of catalyst

Precipitated iron catalysts were prepared using a ferric nitrate solution obtained by dissolving $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ in distilled and deionized water, and then the amount of tetraethyl orthosilicate was added to provide the desired Fe/Si ratio. The mixture was stirred vigorously until the tetraethyl orthosilicate had hydrolyzed. A flow of the tetraethyl orthosilicate and iron nitrate mixture was added to a CSTR precipitation vessel together with a stream of 30% ammonium hydroxide that was added at a rate to maintain a pH of 9.0. By maintaining the slurry pH at 9 and an average residence time of 6 min, a base catalyst material containing iron and silicon was obtained. The slurry from the CSTR was filtered with a vacuum drum filter and then washed twice with deionized water. The final filter cake was dried for 24 h in an oven at 110 °C with flowing air. The catalyst was calcined at 350 °C in an air flow for 4 h. For this study, the Fe/Si catalyst base powder was impregnated with the proper amount of aqueous K_2CO_3 solution to produce the desired composition of Fe/Si/K = 100:5.1:1.25 (atomic ratio). The catalyst was dried at 110 °C overnight following impregnation.

2.2. In situ activation of catalysts

In this study, the potassium promoted iron catalyst batches were pretreated with CO, syngas ($\text{H}_2/\text{CO} = 0.7$) or H_2 at 230–270 °C for 24 h. Carbon monoxide activation was carried out at 1.3 MPa. The CO flowed through a catalyst slurry containing 10% solids in 300 ml of Ethyl flow (a mixture of decene trimers). The reduction of Fe_2O_3 with CO occurs in two steps:



In addition, CO_2 may be formed by the Boudouard reaction:



Previous work in our laboratory showed that approximately 50% to 75% more carbon was present in the CO-activated catalyst mass than was needed to form Fe_5C_2 [28].

H_2 activation was at atmospheric pressure. Similarly, the reduction of Fe_2O_3 with H_2 occurs in two steps:



To investigate the effect of syngas activation pressure on FTS activity, both atmospheric and 1.3 MPa pressure were used. Table 2 shows the detailed activation conditions used in this study.

2.3. Reactor system

A 1-l continuous stirred tank reactor (CSTR) was used in this study. A sintered metal filter was installed to allow removal of wax samples from the catalyst slurry. The wax sample was extracted through the internal filter and collected in a hot trap held at 200 °C. A warm trap (100 °C) and cold trap (0 °C) were used to collect light wax and the water plus oil samples, respectively, by condensing from the vapor phase that was continuously withdrawn from the reactor vapor space.

CO and H_2 mass flow controllers were used to provide a simulated synthesis gas of the desired composition. After the catalyst was activated, syngas was introduced at a rate of 10 NL/h/g Fe. Reaction conditions were 270 °C, 1.3 MPa, $\text{H}_2/\text{CO}=0.7$, and a stirrer speed of 750 rpm.

2.4. Product sampling and analysis

Daily gas, water, oil, light and heavy wax samples were collected and analyzed. A heavy wax sample was taken from the 200 °C hot trap connected to the filter. The vapor phase in the region above the slurry passed continuously to the warm (100 °C) and the cold (0 °C) traps located outside the reactor. The light wax and water mixture was collected daily from the warm trap and an oil plus water sample from the cold trap. Tail gas from the cold trap was analyzed with an online HP Quad Series Micro GC, providing molar compositions of $\text{C}_1\text{--C}_7$ olefins and paraffins as well as for H_2 , CO and CO_2 . Hydrogen and carbon monoxide conversions were calculated based on the gas product GC analysis results and the gas flow measured at the reactor outlet. Hydrogen, carbon

Table 2
Activation conditions for Fe/K/Cu/Si = 100/5/2/1.25 catalyst

Activation gas	Pressure (MPa)	Temperature (°C)	Time (h)	Space velocity (NL/h)
CO	1.3	230/250/270	24	24
Syngas	0.1	230/250/270	24	24
Syngas	1.3	230/250/270	24	24
H_2	0.1	230/250/270	24	24

monoxide and syngas conversion were obtained using the following formula (N = moles of gas):

$$\text{Conversion} = 100\%(N_{\text{in}} - N_{\text{out}})/N_{\text{in}} \quad (8)$$

The oil and light wax samples were mixed before analysis with an Agilent 6890 Series GC. The heavy wax was analyzed with an HP5890 Series II Plus GC while the water sample was analyzed using an HP5890 Series II GC.

3. Results and discussion

3.1. Activity

To investigate the effects of gas type and activation conditions on FTS activity and selectivity, activation with three different activation gases, i.e., H_2 , CO and syngas, were conducted at 230, 250 and 270 °C. Fig. 1 shows the influence of activation with CO, H_2 and syngas at 230 °C on CO conversion. Our previous work showed that for CO activation, essentially the same CO conversion and product selectivity was obtained for activation temperatures in the 230–270 °C range and the same was true for activation pressures in the range of 0.1–1.3 MPa [9]. Activation with CO was carried out at 1.3 MPa while both H_2 and syngas activations were conducted at atmospheric pressure in this study. In general, higher CO conversion is obtained when activations with H_2 or syngas are conducted at 0.1 MPa than at reaction pressure. The highest initial CO conversion was

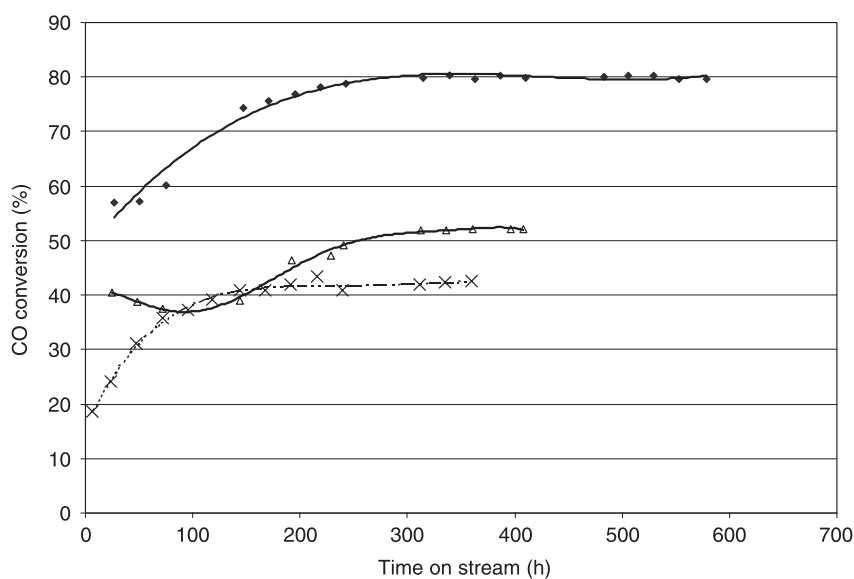


Fig. 1. Influence of activation gas type on CO conversion (activated at 230 °C; ◆, CO activation, 1.3 MPa; ×, hydrogen activation, 0.1 MPa; △, syngas activation, 0.1 MPa; FTS conditions: 270 °C, 1.3 MPa).

obtained from CO activated catalyst while the lowest CO conversion was produced by the H₂ activated catalyst. At 230 °C, CO activation yielded a stable CO conversion of 80% after going through a 280-h induction period, in which CO conversion increased monotonously from 50%. Similar to the trend for CO activation, hydrogen activation showed an induction period of 120 h before reaching a stable CO conversion of 42%. Syngas activation, however, showed an initial decrease in CO conversion from above 40% to 37% before increasing to a stable level of 52% after 320 h of time on stream. These results show that activation gas type has an important influence on FTS activity. A CO conversion of 42% was obtained for the H₂ activated catalyst while CO activation produced a CO conversion of 80%, almost twice as high as that from H₂ activated catalyst.

Fig. 2 shows the results for activation at 250 °C with the three different gases on CO conversion. Overall CO conversion for the CO activated catalyst was higher than both H₂ and syngas activated catalysts. The CO activated catalyst yielded a stable CO conversion of 80%; the CO conversion for the syngas activated catalyst was 62% and the H₂ activated catalyst was 43%. Although a stable CO conversion of 80% was produced for both 230 and 250 °C activation temperatures when CO was used as activation gas, activation at 250 °C yielded a peak CO conversion of 85% after a 200-h induction period. Hydrogen activation generated a peak CO conversion of 58% after a 100-h induction period before reaching a stable CO conversion of 42%. Increase in activation temperature did not change the stabilized CO conversion of both H₂ and CO activated catalysts, although a higher initial activity was observed for the catalysts activated at 250 °C than that at 230 °C. Syngas activated catalyst, however, showed a significant improvement in CO conversion as the activation temperature was increased from 230 to 250 °C. The initial CO conversion increased from 45% and the stabilized conversion increased from 52% to 62% when the activation temperature was increased from 230 to 250 °C.

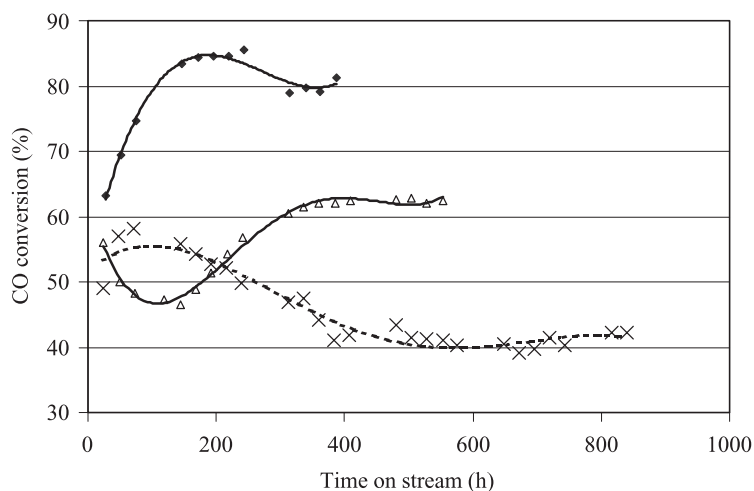


Fig. 2. Influence of activation gas type on CO conversion (activated at 250 °C; ◆, CO activation, 1.3 MPa; ×, hydrogen activation, 0.1 MPa; △, syngas activation, 0.1 MPa; FTS conditions: 270 °C, 1.3 MPa).

Similar to the results shown in Figs. 1 and 2, carbon monoxide activated catalyst yielded the highest activity at the 270 °C activation temperature, as indicated by Fig. 3. A stabilized CO conversion of 84% was obtained at only 200 h of time on stream and deactivated slightly after 600 h of time on stream. Syngas activation yielded a stabilized CO conversion of 80% at 250 h of time on stream while H₂ activation generated a material with a CO conversion of 59% at 320 h of time on stream. An increase in activation temperature from 250 to 270 °C significantly improved the stable FTS activity regardless of the gas type used in the activation process, suggesting that higher temperature enhanced the activation process.

As claimed by other researchers, activation conditions can also influence the FTS activity [1,3,25]. Fig. 4 shows the effect of activation pressure on FTS activity of the syngas activated catalyst. For atmospheric pressure activation, a CO conversion of 42%, 62% and 80% was achieved at 230, 250 and 270 °C, respectively, while a low CO conversion of less than 15% was obtained at all temperatures under 1.3 MPa pressure. These results indicate that pressure had a detrimental effect on FTS activity following syngas activation. Fig. 5 shows the effect on CO conversion of temperature for H₂ activation from 230 to 250 °C; the peak CO conversion during the initial induction period increased from 42% to 58%, while the stabilized CO conversion was the same for both temperatures. After 400 h of time on stream, a stabilized CO conversion of 40% was obtained at both 230 and 250 °C. For activation at 270 °C, however, both peak CO conversion and the stabilized CO conversion were higher. A 76% peak conversion and a 55% steady-state CO conversion resulted from activation at 270 °C. Thus a higher activation temperature is desirable for hydrogen activated iron catalyst to obtain a higher CO conversion, compared to activation at 230 or 250 °C.

Higher activation temperature can give a higher hydrocarbon production rate when CO or syngas was used as activation gas, as shown in Fig. 6. When CO was used as activation gas, the hydrocarbon production rate remained unchanged when the activation temperature

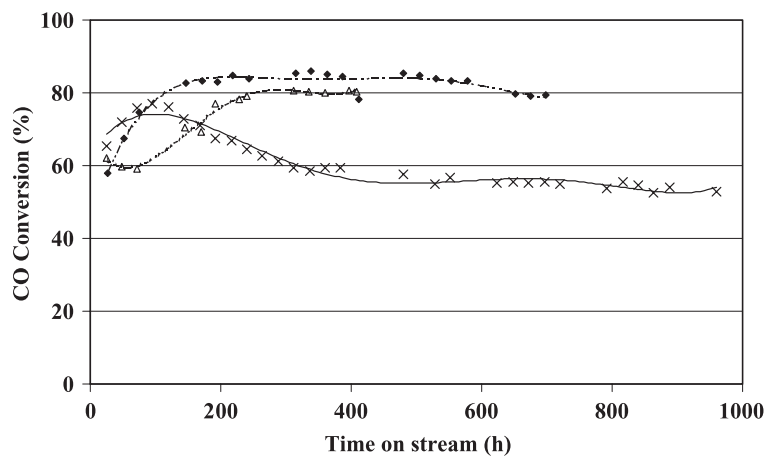


Fig. 3. Influence of activation gas type on CO conversion (activated at 270 °C; ◆, CO activation, 1.3 MPa; ×, hydrogen activation, 0.1 MPa; △, syngas activation, 0.1 MPa; FTS conditions: 270 °C, 1.3 MPa).

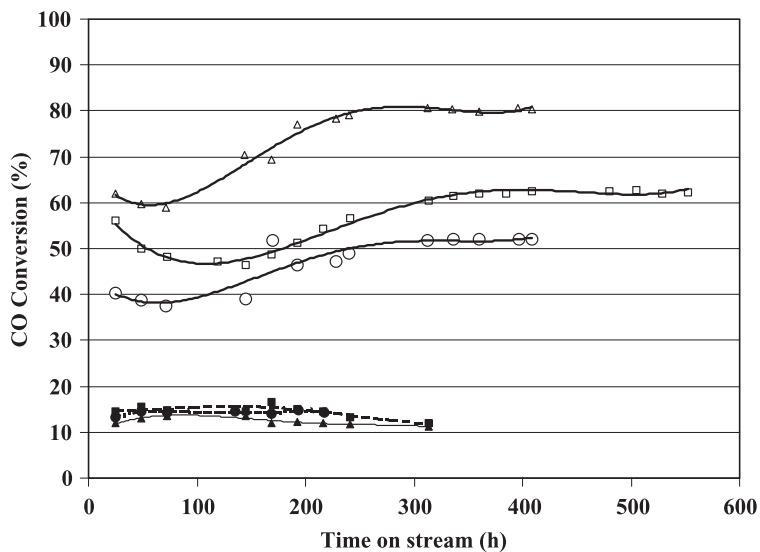


Fig. 4. Influence of syngas activation pressure and temperature on CO conversion (●, syngas, 230 °C, 1.3 MPa; ■, syngas, 250 °C, 1.3 MPa; ▲, syngas, 270 °C, 1.3 MPa; ○, syngas, 230 °C, 0.1 MPa; □, syngas, 250 °C, 0.1 MPa; △, syngas, 270 °C, 0.1 MPa) (FTS conditions: 270 °C, 1.3 MPa).

increased from 230 to 250 °C, then increased from 1.67 to 1.75 g/h g Fe following activation at 270 °C. Activation temperature with syngas showed a greater effect on hydrocarbon production rate than for CO activation. An increase in activation temperature from 230 to 250 °C improved the hydrocarbon rate from 1.16 to 1.29 g/h g Fe, and an increase in activation temperature to 270 °C increased the hydrocarbon production rate to

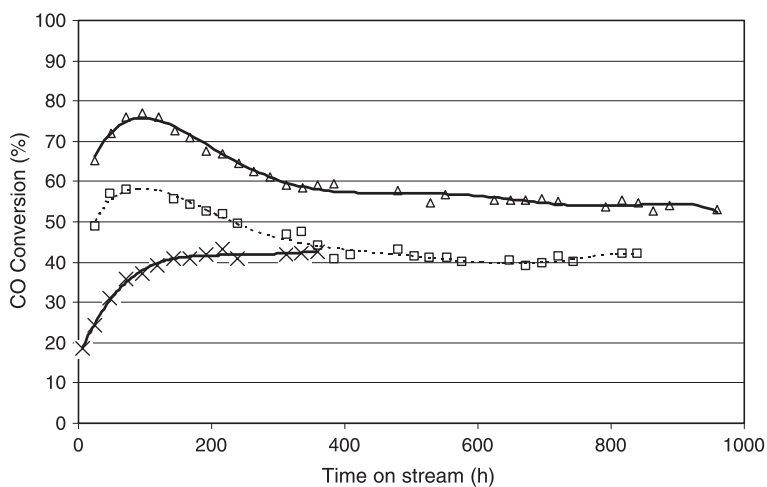


Fig. 5. Influence of hydrogen activation temperature on CO conversion (×, hydrogen, 230 °C, 0.1 MPa; □, hydrogen, 250 °C, 0.1 MPa; △, hydrogen, 270 °C) (FTS conditions: 270 °C, 1.3 MPa).

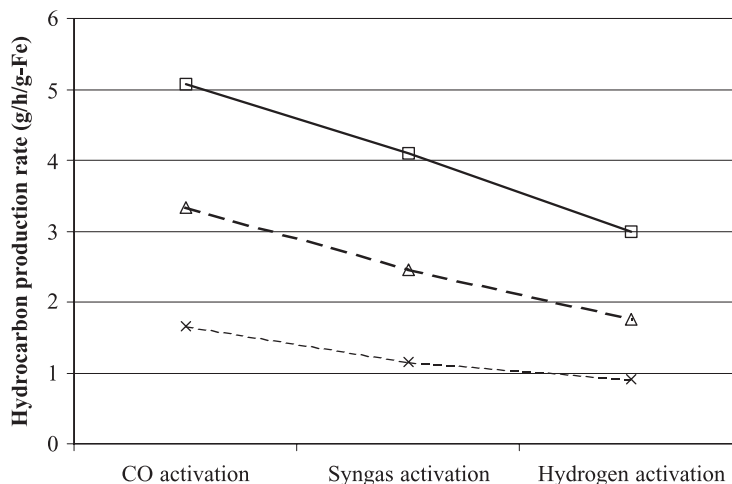


Fig. 6. Influence of activation gas on hydrocarbon production rate (FTS conditions: 270 °C, 1.3 MPa) (×, activated at 230 °C; Δ, activated at 250 °C; □, activated at 270 °C).

1.64. When H₂ was used as activation gas, a slight decrease (from 0.91 to 0.84 g/h g Fe) for the hydrocarbon production may have resulted from increasing the temperature from 230 to 250 °C; the rate increased to 1.24 g/h g Fe when the temperature was increased to 270 °C. Carbon monoxide activation yielded a higher overall hydrocarbon rate than both syngas and H₂, regardless of the activation temperature.

3.2. Selectivity

Methane selectivity was defined as:

$$\text{CH}_4 = \frac{\text{rate of CH}_4}{\text{rate of FTS}} (100) = \frac{\text{CH}_4 \text{ produced}}{(\text{CO reacted} - \text{CO}_2 \text{ produced})} (100)$$

All hydrocarbon selectivity data were obtained after CO conversion reached a stabilized level.

Fig. 7 gives the methane selectivity for FTS with three different activation gases at 230, 250 and 270 °C. With CO activation, methane selectivity did not depend on activation temperature. The best overall methane selectivity was also obtained with CO activation as only about 8% methane was produced with this activation gas. Syngas activation generated 12.2% methane selectivity at 250 °C but was about 8% at 250 and 270 °C. Hydrogen activation yielded a methane selectivity of 8%, 9% and 10% at 230, 270 and 250 °C, respectively. The lowest methane selectivity (7.6%) was obtained at 250 °C with CO activation and 270 °C with syngas activation while the highest methane selectivity (12.2%) was produced with syngas activation at 250 °C.

Carbon monoxide activation yielded a C₂–C₄ fraction similar to syngas activation. Activation temperature had a slight effect on C₂–C₄ fraction with CO or syngas activation, as indicated in Fig. 8. An increase in activation temperature from 230 to 250 and 270 °C

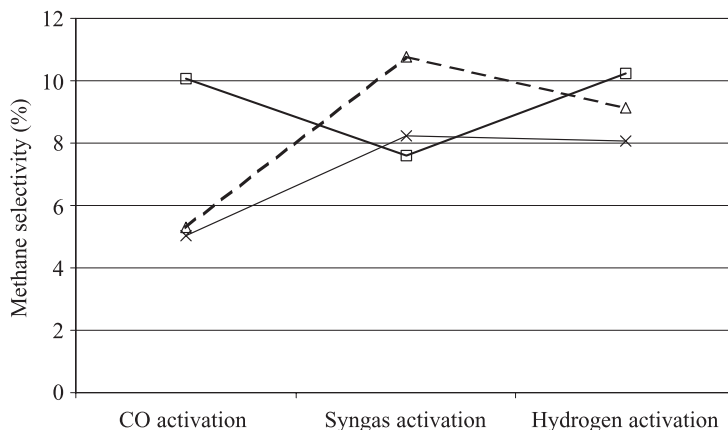


Fig. 7. Influence of activation gas on methane selectivity (FTS conditions: 270 °C, 1.3 MPa) (x, activated at 230 °C; Δ, activated at 250 °C; □, activated at 270 °C).

resulted in a slight increase from 29.1% to 30.4% and 30.8% with CO activation, and a slight decrease with syngas activation from 31.1% to 30.1% and 29.6%, respectively. With H₂ activation, the C₂–C₄ fraction exhibited little change when the activation temperature increased from 230 to 250 °C. When the activation temperature increased from 250 to 270 °C with H₂, however, a greater change in C₂–C₄ fraction from 26.4% to 31.2% was observed.

Fig. 9 shows the effect of activation gas and activation temperature on the gasoline fraction (C₅–C₁₁) yield. Hydrogen activation yielded the highest C₅–C₁₁ fraction while syngas activation yielded the lowest. The highest gasoline fraction of 47.8% was produced

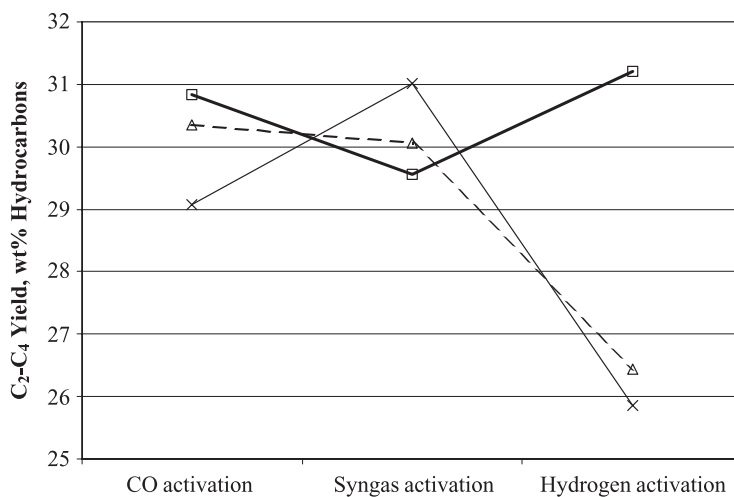


Fig. 8. Influence of activation gas on C₂–C₄ yield (FTS conditions: 270 °C, 1.3 MPa) (x, activated at 230 °C; Δ, activated at 250 °C; □, activated at 270 °C).

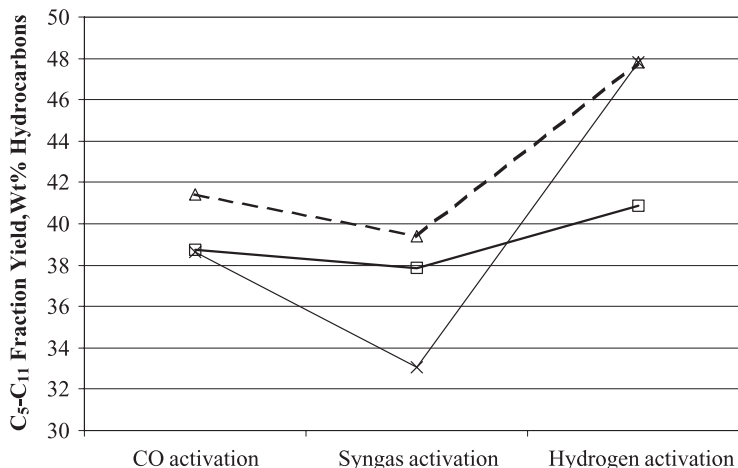


Fig. 9. Influence of activation gas on C_5 – C_{11} fraction yield (FTS conditions: 270 °C, 1.3 MPa) (×, activated at 230 °C; Δ, activated at 250 °C; □, activated at 270 °C).

at both 230 and 250 °C activations with H_2 . At 270 °C, H_2 activation yielded a gasoline fraction of 40.9%. Carbon monoxide activation resulted in a gasoline yield of 39% to 41% over the activation temperature range. Syngas activation produced a C_5 – C_{11} fraction similar to CO activation at 250 and 270 °C. The lowest gasoline fraction was obtained at 230 °C when syngas was used as activation gas. Fig. 9 also shows that activation temperature did not significantly affect the C_5 – C_{11} fraction as activation temperature increased from 230 to 270 °C.

Effect of activation gas type and activation temperature on diesel fraction (C_{12} – C_{18}) from FTS over iron catalyst is shown in Fig. 10. Higher diesel fraction was obtained from

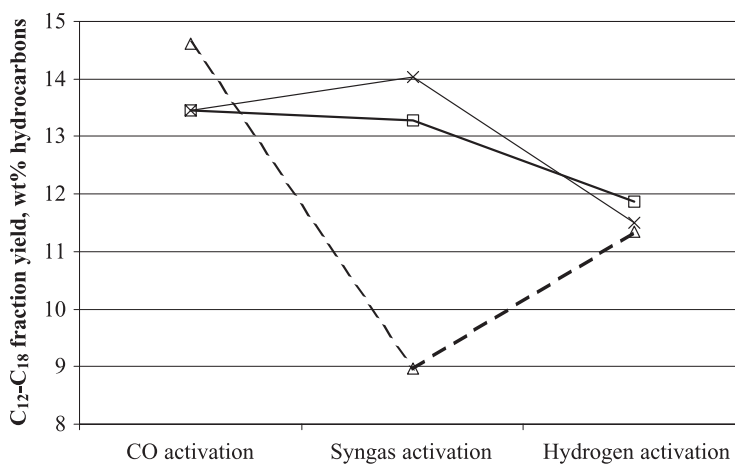


Fig. 10. Influence of activation gas on C_{12} – C_{18} fraction yield (FTS conditions: 270 °C, 1.3 MPa) (×, activated at 230 °C; Δ, activated at 250 °C; □, activated at 270 °C).

CO and syngas activation than from H_2 activation. Activation temperature showed little effect on diesel yield with CO and H_2 activation. Increase in activation temperature resulted in a diesel fraction change from 13.5% to 14.6% with CO activation and 11.3% to 11.9% with H_2 activation. Thus, the syngas activation at 250 °C result is likely due to experimental error.

Although CO activation yielded the highest diesel fraction, syngas activation produced the highest C_{19+} fraction (Fig. 11). Samples from different runs were collected after CO conversion was stabilized. The highest C_{19+} fraction of 10% was obtained at 230 °C with syngas activation and the lowest C_{19+} fraction of 3.1–3.4% at 250 °C with CO or H_2 activation. Regardless of the activation gas type, activation temperature significantly affected the C_{19+} fraction. As shown in Fig. 11, activation at 250 °C yielded a lower C_{19+} fraction than at 250 and at 270 °C. With CO activation, C_{19+} fraction changed from 7.4% to 3.1% and 4.9% as activation temperature increased from 230 to 250 and 270 °C, respectively. Hydrogen activation produced the lowest overall C_{19+} fraction. As shown in the figure, only a 3–5% C_{19+} yield was obtained when H_2 was used as activation gas. However, the catalyst has a selectivity that falls in the low to medium alpha range and the low C_{19} will cause a large experimental error.

Fischer–Tropsch synthesis can produce some valuable chemicals, such as straight chain paraffins, α -olefins, internal olefins and, branched paraffins and olefins. FTS reaction conditions, catalyst type, promoter type and contents can have an important influence on product selectivity [1]. Activation conditions and activation gas type can also affect the product selectivity [1,9,25]. The highest C_2 – C_{20} olefin fraction was obtained from H_2 activation and the lowest from CO activation, as shown in Figs. 12–14. In making this comparison, it should be recognized that the olefin fraction depends upon CO conversion, and the different conversion levels are responsible for part of the differences.

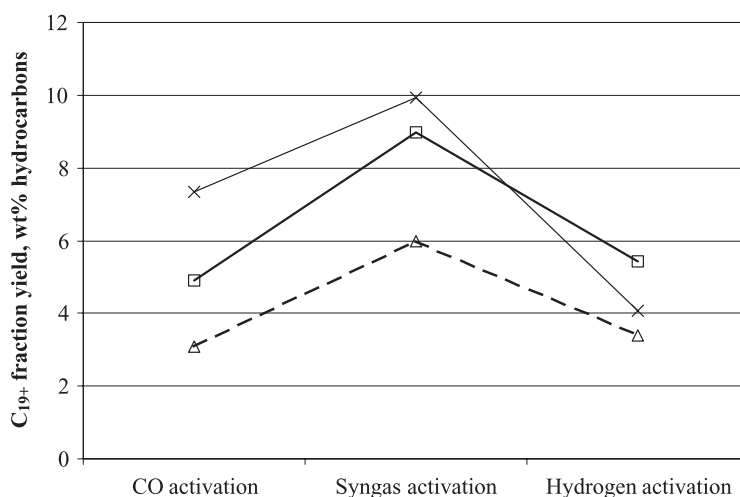


Fig. 11. Influence of activation gas on C_{19+} fraction yield (FTS conditions: 270 °C, 1.3 MPa) (x, activated at 230 °C; Δ, activated at 250 °C; □, activated at 270 °C).

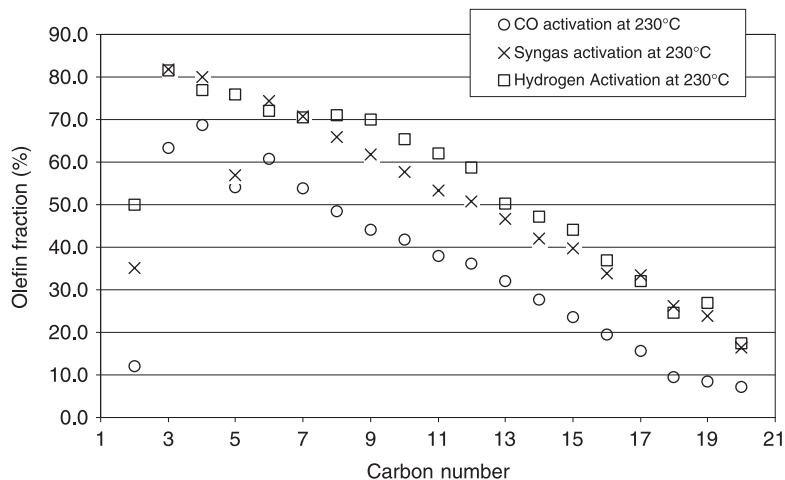


Fig. 12. Influence of activation gas on olefin fraction by carbon number (FTS conditions: 270 °C, 1.3 MPa).

With the exception of ethylene, all olefin ratios decreased linearly with increasing carbon, regardless of the activation temperature. Ethylene was believed to possess a rapid readsorption ability in the FTS system [29–31]. Explanations for the low ethylene are incorporation of ethylene into the growing chain [32,33], hydrogenolysis of ethylene [30] and hydrogenation of ethylene [31,34,35]. Figs. 12–14 also shows a deviation from the linear olefin ratio distribution for pentene with CO and syngas activation regardless of the activation temperature; this is likely due to the inclusion of a second component in the g.c. peak assigned to pentane.

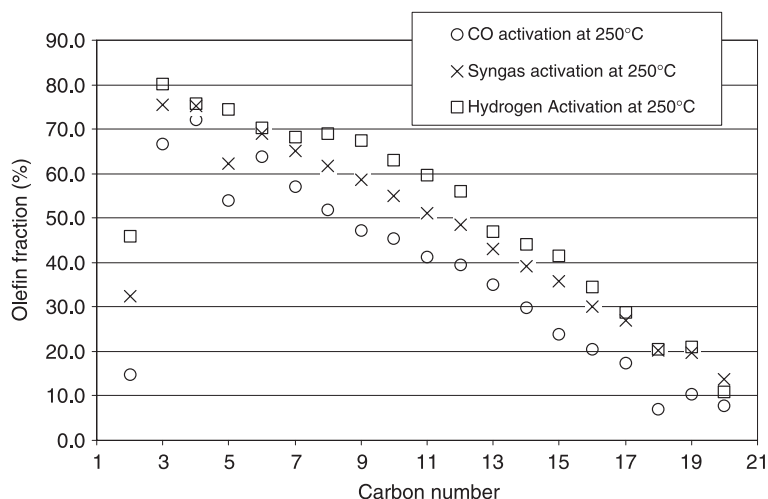


Fig. 13. Influence of activation gas on olefin ratio (FTS conditions: 270 °C, 1.3 MPa).

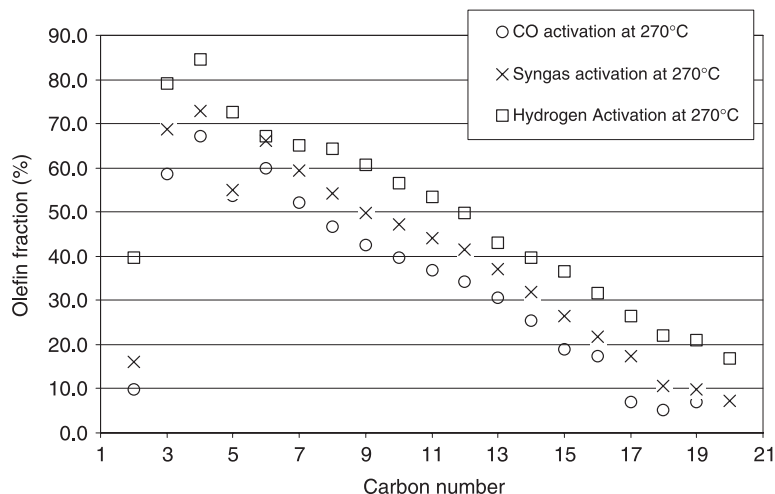


Fig. 14. Influence of activation gas on olefin ratio (FTS conditions: 270 °C, 1.3 MPa).

3.3. Water gas shift

Water gas shift (WGS) is an important reaction in an FTS process when an iron catalyst is utilized. In this reaction, CO reacts with water, which is generated from FTS and produces CO_2 and H_2 . WGS produces the majority of the CO_2 from FTS system, as given by Eq. (2). Fig. 15 shows the influence of activation temperature and activation gas type on CO_2 selectivity. This indicates that activation temperature did not affect the CO_2 selectivity with CO activation. With syngas and H_2 activation, however, CO_2 selectivity was affected

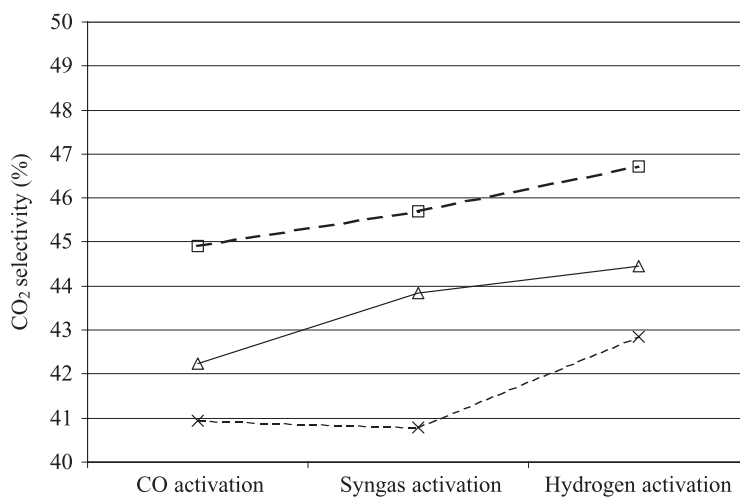


Fig. 15. Influence of activation gas on CO_2 selectivity (FTS conditions: 270 °C, 1.2 MPa) ($\times$, activated at 230 °C; Δ , activated at 250 °C; $\square$, activated at 270 °C).

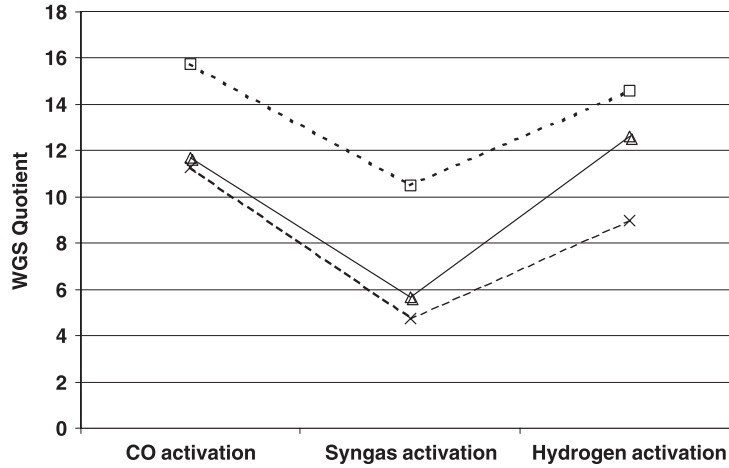


Fig. 16. Influence of activation gas on the WGS quotient, $Q_{\text{WGS}} = P[\text{CO}_2]P[\text{H}_2]/P[\text{H}_2\text{O}]$ (FTS conditions: 1.3 MPa, 270 °C) (x, activated at 230 °C; Δ, activated at 250 °C; □, activated at 270 °C).

appreciably by activation temperature. Carbon monoxide activation generated a CO_2 selectivity of 45% regardless of the activation temperature. As activation temperature increased from 230 to 250 °C, CO_2 selectivity increased from 41% to 45% with syngas activation, and from 43% to 47% with H_2 activation. Further increase in activation temperature slightly raised the CO_2 from 45% to 46% with syngas activation, but lowered the CO_2 selectivity from 47% to 44.5% with H_2 activation. The highest CO_2 of 47% was produced with H_2 activation at 250 °C while the lowest of 41% obtained with syngas activation at 230 °C.

Water gas shift quotient, as given by Eq. (9), was used to show the extent of the WGS reaction:

$$Q_{\text{WGS}} = \frac{P[\text{CO}]P[\text{H}_2\text{O}]}{P[\text{CO}_2]P[\text{H}_2]} \quad (9)$$

Fig. 16 shows the effect of activation gas type and activation temperature on water gas shift quotient. With syngas and H_2 activation, water gas shift quotient increased monotonously as activation temperature increased from 230 to 270 °C. Increase in activation temperature from 230 to 250 °C shows a slight decrease in water gas shift quotient from 12.6 to 11, but an increase of water gas shift quotient from 11 to 15.8 as activation temperature increased from 250 to 270 °C. The highest water gas shift quotient was obtained with CO activation at 270 °C while the lowest obtained with syngas activation at 230 to 250 °C.

4. Conclusions

The highest CO conversion is obtained for a CO activated catalyst, regardless of the activation temperature. Although different initial CO conversions are obtained at 250 and

270 °C, stabilized CO conversion (80%) is independent of the activation temperature when CO is used as activation gas. In contrast, hydrogen activation yielded the lowest CO conversion among the three activation gases. Syngas activation temperature increased the CO conversion as activation temperature increased from 230 to 270 °C. The results indicate that activation temperature and activation gas type produce a different influence on FTS activity. Syngas pretreatment shows that FTS activity depends on the activation pressure. For high pressure, a low CO conversion is obtained, regardless of the activation temperature. At atmospheric pressure, however, FTS activity is noticeably increased and activation temperature shows an influence on CO conversion.

In addition to FTS activity, CO activation shows the best overall hydrocarbon productivity and the best methane selectivity. The lowest hydrocarbon rate is obtained from H₂ activation. Syngas activation generates the lowest water gas shift activity while CO activation produces the highest. It is also found that activation temperature impacts the WGS activity. The higher WGS activity from CO activated catalyst results in a lower H₂/CO usage as the WGS consumes CO and generates additional H₂. Activation temperature has little effect on CO₂ selectivity and H₂/CO usage with CO activation. With syngas activation, however, temperature has a negative effect on H₂/CO usage and a positive effect on CO₂ selectivity.

Syngas and CO activation produces a similar gas (C₂–C₄) and gasoline (C₅–C₁₁) fractions, regardless of the activation temperature. Only H₂ activation at 230–250 °C shows a noticeably lower C₂–C₄ higher C₅–C₁₁ fractions than with syngas and CO activation. While CO activation produces a slightly higher diesel (C₁₂–C₁₈) fraction, syngas yields a higher C₁₉₊ fraction than other activation gases. Activation temperature has little effect on the formation of olefin (C₂–C₂₀) and the olefin ratio decreases in the order of H₂>syngas>CO activation.

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References

- [1] M.E. Dry, in: *Catalysis: Science and Technology*, vol. 1, Springer Verlag, New York, NY, 1981, pp. 159–255.
- [2] R.J. O'Brien, L. Xu, R.L. Spicer, B.H. Davis, *Symposium on Syngas Conversion to High Value Chemicals*, 252–253, 211th ACS Meeting, New Orleans, March 24–29, 1996.
- [3] R.A. Dector, A.T. Bell, *J. Catal.* 97 (1986) 121.
- [4] D.B. Bukur, D. Mukesh, S.A. Patel, *Ind. Eng. Chem. Res.* 29 (1990) 194.
- [5] M.E. Dry, T. Shingles, L.J. Boshoff, G.J. Oosthuizen, *J. Catal.* 15 (1969) 190.
- [6] J. Benziger, R.J. Madix, *Surf. Sci.* 94 (1980) 119.
- [7] H. Arakawa, A.T. Bell, *IEC Process Des. Dev.* 22 (1993) 97.
- [8] M. Luo, B.H. Davis, *Studies in Surface Science and Catalysis*, in: J.J. Spivey, G.W. Roberts, B.H. Davis (Eds.), *Catalyst Deactivation 2001*, vol. 139, Elsevier Science, Amsterdam, The Netherlands, 2001, pp. 122–140.
- [9] R.J. O'Brien, L. Xu, R.L. Spicer, B.H. Davis, *Energy Fuels* 10 (1996) 921.

- [10] R.B. Anderson, Catalysis for Fischer–Tropsch synthesis, in: P.H. Emmertt (Ed.), *Catalysis*, vol. 4, Reinhold Publishing, New York, 1958, p. 29.
- [11] J.P. Baltrus, J.R. Diehl, M.A. McDonald, M.F. Zarochak, *Appl. Catal.* 48 (1989) 199.
- [12] A.G. Sault, *J. Catal.* 140 (1993) 121.
- [13] V.U.S. Rao, G.J. Steigel, G.J. Cinquegrane, R.D. Srivastava, *Fuel Process. Technol.* 30 (1992) 83.
- [14] M.E. Dry, in: B.E. Leach (Ed.), *Applied Industrial Catalysis*, vol. 2, Academic Press, New York, 1983, p. 167.
- [15] C.D. Frohning, in: J. Falbe (Ed.), *New Synthesis with Carbon Monoxide*, Springer-Verlag, New York, 1980, p. 353.
- [16] C. Li, “Effect of potassium and copper promoters on reduction behavior of precipitated iron catalysts”, PhD Thesis, Texas A&M University, 1988.
- [17] J.L. Rankins, C.H. Bartholomew, *J. Catal.* 100 (1986) 533.
- [18] J.-A. Dalmon, R. Dutartre, G.A.C.R. Martin, *Acad. Sci., Ser. C* 287 (1978) 557.
- [19] A.G. Sault, *J. Catal.* 140 (1993) 136.
- [20] D.J. Dwyer, J.H. Hardenburg, *Appl. Surf. Sci.* 19 (1984) 14.
- [21] H.P. Bonzel, H.J. Kreh, *Surf. Sci.* 109 (1981) 1527.
- [22] D.A. Wesner, F.P. Coenen, H.P. Bonzel, *Langmuir* 1 (1985) 478.
- [23] D.B. Bukur, L. Nowichi, X. Lang, *Energy Fuels* 9 (1995) 620.
- [24] K.R.P.M. Rao, F.E. Huggins, V. Mahajan, G.P. Huffman, B.L. Rao, V.U.S. Bhatt, D.B. Bukur, B.H. Davis, R.J. O’Brien, *Top. Catal.* 2 (1995) 71.
- [25] D.B. Bukur, M. Koranne, X. Lang, K.R.P.M. Rao, G.P. Huffman, *Appl. Catal. A Gen.* 126 (1995) 85.
- [26] E.S. Lox, G.B. Marin, E. De Graeve, P. Bussiere, *Appl. Catal. A Gen.* 40 (1988) 197.
- [27] R.J. O’Brien, Y. Zhang, H.H. Hamdeh, B.H. Davis, *Preprints*, vol. 44(1), ACS, Div. of Petroleum Chemistry, Anaheim, CA, 1999, March 21–25, pp. 100–102.
- [28] A.P. Raje, R.J. O’Brien, B.H. Davis, *J. Catal.* 180 (1998) 36.
- [29] E. Iglesia, S.C. Reyes, R.J. Madon, *J. Catal.* 129 (1991) 238.
- [30] T. Komaya, A.T. Bell, *J. Catal.* 146 (1994) 237.
- [31] D.S. Jordon, A.T. Bell, *J. Phys. Chem.* 90 (1986) 4797.
- [32] S. Novak, R.J. Madon, H. Shul, *J. Catal.* 77 (1982) 141.
- [33] S. Novak, R.J. Madon, H. Shul, *J. Phys. Chem.* 74 (1981) 6083.
- [34] E.W. Kuiper, I.H. Vinkenburg, H. Oosterbeek, *J. Catal.* 152 (1995) 137.
- [35] E.W. Kuiper, C. Scheper, J.H. Wilson, H. Oosterbeek, *J. Catal.* 158 (1996) 288.